Feature Reference

AI for Limit Order Execution - static order features, contemporaneous LOB state, recent event flow, and 50-snapshot history

Single-Symbol Synthetic Limit-Order-Book Dataset

This document defines the variables supplied in **AI_Order_Execution.xlsx** and the raw historical fields described by its **JSON_Schema** sheet. It also defines a reproducible set of summary features that can be constructed from the 50-snapshot history.

The dataset contains synthetic teaching data for one symbol, AAPL. Definitions in this document describe the supplied dataset and should not be interpreted as empirical claims about real AAPL trading.

1. Conventions

Observation time

t0 is the order-level observation time. Every predictor in the **Orders** sheet is measured at or before **t0**. Outcomes are observed after **t0**.

Book levels

- Level 1 is the best quote.
- Bid prices decrease from L1 to L5.
- Ask prices increase from L1 to L5.
- **tick_size** is USD 0.01 for every row.

Order-side orientation

Resting order	Same side	Opposite side	Aggressive volume against the order
Buy	Bid	Ask	Aggressive sell volume consuming bids.
Sell	Ask	Bid	Aggressive buy volume consuming asks.

Time windows

Fields ending in **_10s** summarize the 10-second interval immediately preceding **t0**. **volatility_60s_bps** summarizes the 60-second interval ending at **t0**.

2. Order context and queue state

Field	Type / unit	Definition
side	Category	Direction of the resting order. **buy** orders rest on the bid side; **sell** orders rest on the ask side.
seconds_from_open	Decimal / seconds	Seconds elapsed between the 09:30 market open and **t0**. Range: 0-23,400.
session_phase	Category	Intraday phase at **t0**: **opening** for the first 30 minutes, **continuous** for the middle session, and **closing** for the final hour.
order_age_s	Decimal / seconds	Seconds elapsed between the order's original submission and **t0**.
limit_price	Decimal / USD per share	Fixed limit price of the resting order at **t0**. The value is aligned to **tick_size**.
order_size	Integer / shares	Original submitted order quantity. Range: 1-5,000 shares.
price_distance_ticks	Integer / ticks	Distance from the order price to the same-side best quote at **t0**. For a buy order: **(bid_price_l1 - limit_price) / tick_size**. For a sell order: **(limit_price - ask_price_l1) / tick_size**. Zero means that the order rests at the best quote. Range: 0-4.
quantity_filled_before_t0	Integer / shares	Quantity executed between original submission and **t0**. It is strictly smaller than **order_size** because the order remains active at **t0**.
queue_ahead_qty	Integer / shares	Displayed quantity ahead of the studied order at the same limit price at **t0**.
same_price_total_qty	Integer / shares	Total displayed quantity at the studied order's price, including quantity ahead, the order's remaining displayed quantity, and quantity behind it.
normalized_queue_ahead	Decimal / ratio	Relative queue position: **queue_ahead_qty / same_price_total_qty**. Range: **[0, 1)**.

3. Five-level limit order book at t0

Bid and ask prices

Fields	Type / unit	Definition
bid_price_l1 to **bid_price_l5**	Decimal / USD per share	Displayed bid prices at levels 1-5. L1 is the highest bid; deeper bid prices decrease in tick increments.
ask_price_l1 to **ask_price_l5**	Decimal / USD per share	Displayed ask prices at levels 1-5. L1 is the lowest ask; deeper ask prices increase in tick increments.

Bid and ask volumes

Fields	Type / unit	Definition
bid_volume_l1 to **bid_volume_l5**	Integer / shares	Displayed bid quantity at each of the five price levels.
ask_volume_l1 to **ask_volume_l5**	Integer / shares	Displayed ask quantity at each of the five price levels.

The raw five-level book therefore contains 20 columns: five bid prices, five bid volumes, five ask prices, and five ask volumes.

4. Derived LOB state at t0

Define:

- **P_bid = bid_price_l1**
- **P_ask = ask_price_l1**
- **V_bid = bid_volume_l1**
- **V_ask = ask_volume_l1**
- **D_bid = sum(bid_volume_l1, ..., bid_volume_l5)**
- **D_ask = sum(ask_volume_l1, ..., ask_volume_l5)**

Field	Type / unit	Exact definition
mid_price	Decimal / USD per share	Arithmetic midpoint: **(P_bid + P_ask) / 2**.
spread_ticks	Integer / ticks	Best-quote spread in ticks: **(P_ask - P_bid) / tick_size**. Range: 1-4.
relative_spread_bps	Decimal / basis points	Best-quote spread relative to the midpoint: **10,000 * (P_ask - P_bid) / mid_price**.
top_level_imbalance	Decimal / ratio	L1 bid-ask volume imbalance: **(V_bid - V_ask) / (V_bid + V_ask)**. Range: **[-1, 1]**. This field is not oriented by the studied order's side.
same_side_depth_5	Integer / shares	Five-level depth on the studied order's side. It equals **D_bid** for a buy order and **D_ask** for a sell order.
opposite_side_depth_5	Integer / shares	Five-level depth opposite the studied order. It equals **D_ask** for a buy order and **D_bid** for a sell order.
five_level_side_imbalance	Decimal / ratio	Side-oriented five-level imbalance: **(same_side_depth_5 - opposite_side_depth_5) / (same_side_depth_5 + opposite_side_depth_5)**. Range: **[-1, 1]**.
microprice	Decimal / USD per share	Top-level volume-weighted price: **(V_bid *P_ask* + *V_ask* P_bid) / (V_bid + V_ask)**. It lies between the best bid and best ask.
volatility_60s_bps	Decimal / basis points	Synthetic realized-volatility signal over the 60 seconds ending at **t0**. The workbook does not specify the sampling frequency, return convention, or annualization rule used to generate this supplied field.

5. Recent order flow before t0

Submissions, cancellations, and modifications

Field	Type / unit	Definition
same_side_submissions_10s	Integer / messages	New limit-order submissions on the studied order's side during the 10 seconds before **t0**.
opposite_side_submissions_10s	Integer / messages	New limit-order submissions on the side opposite the studied order during the 10 seconds before **t0**.
same_side_cancellations_10s	Integer / messages	Cancellation messages on the studied order's side during the 10 seconds before **t0**. This is an aggregate side-level count; it does not state that each cancellation occurred ahead of the studied order or at its price.
opposite_side_cancellations_10s	Integer / messages	Cancellation messages on the side opposite the studied order during the 10 seconds before **t0**.
modifications_10s	Integer / messages	Order-modification messages on both sides during the 10 seconds before **t0**.

Trades and activity

Field	Type / unit	Definition
aggressive_trade_volume_against_order_10s	Integer / shares	Aggressive volume that consumed resting liquidity on the studied order's side during the 10 seconds before **t0**. For a resting buy order this is aggressive sell volume; for a resting sell order this is aggressive buy volume.
aggressive_trade_volume_with_order_side_10s	Integer / shares	Aggressive volume in the same trading direction as the studied order during the 10 seconds before **t0**. For a resting buy order this is aggressive buy volume consuming asks; for a resting sell order this is aggressive sell volume consuming bids.
trade_count_10s	Integer / trades	Number of trades during the 10 seconds before **t0**.
trade_intensity_per_s	Decimal / trades per second	**trade_count_10s / 10**. This is a deterministic rescaling of **trade_count_10s**.
same_side_cancel_intensity_per_s	Decimal / cancellations per second	**same_side_cancellations_10s / 10**. This is a deterministic rescaling of **same_side_cancellations_10s**.
time_since_last_trade_ms	Decimal / milliseconds	Milliseconds elapsed between the most recent trade and **t0**.
market_activity_group	Category	Activity regime derived from recent order-flow intensity: **low**, **medium**, or **high**. The workbook does not specify the numerical thresholds used to construct the categories.

6. Historical JSON structure

The history data contain one object per order and join to the **Orders** sheet using **order_id**. Each order has exactly 50 snapshots in chronological order, all strictly earlier than the order's original submission time.

Order-level JSON fields

JSON field	Type / unit	Definition
orders[].order_id	Text	Join key corresponding to **Orders.order_id**.
orders[].symbol	Category	**AAPL** for every order.
orders[].order_type	Category	Synthetic generation mechanism: **observed** or **hypothetical_pegged**. This is metadata, not an order-book state variable.
orders[].submission_timestamp	ISO-8601 timestamp	Original order submission time. It precedes **observation_timestamp** by **order_age_s**.
orders[].lookback_count	Integer	Number of historical snapshots. Always 50.
orders[].history	Array of 50 objects	Chronological sequence of pre-submission LOB snapshots.

Snapshot fields

JSON field	Type / unit	Definition
sequence_step	Integer	Position within the history. Step 1 is oldest; step 50 is closest to submission.
event_timestamp	ISO-8601 timestamp	Timestamp of the LOB event represented by the snapshot. It is strictly earlier than **submission_timestamp**.
milliseconds_before_submission	Integer / milliseconds	Negative time offset from submission. The closest snapshot has value -1 ms.
event_type	Category	Event producing the snapshot: **limit_submission**, **cancellation**, **market_trade**, or **modification**.
event_side	Category	Side changed by the event: **bid** or **ask**.
event_level	Integer	Affected LOB level, from 1 to 5.
bid_prices	Array[5] / USD per share	Bid prices from L1 to L5.
bid_volumes	Array[5] / shares	Displayed bid quantities from L1 to L5.
ask_prices	Array[5] / USD per share	Ask prices from L1 to L5.
ask_volumes	Array[5] / shares	Displayed ask quantities from L1 to L5.
mid_price	Decimal / USD per share	Mean of the best bid and best ask in the snapshot.
spread_ticks	Integer / ticks	Best-quote spread in the snapshot.
top_level_imbalance	Decimal / ratio	L1 bid-ask volume imbalance for the snapshot.
five_level_bid_ask_imbalance	Decimal / ratio	**(total bid depth - total ask depth) / (total bid depth + total ask depth)** across five levels. This raw JSON field is bid-ask oriented, not studied-order-side oriented.
microprice	Decimal / USD per share	Top-level volume-weighted microprice for the snapshot.
volatility_60s_bps	Decimal / basis points	Synthetic recent realized-volatility signal recorded at the snapshot.

Snapshot change fields

Each change field is measured relative to the preceding snapshot. All change arrays contain zeros at **sequence_step = 1** because no earlier snapshot exists in the stored history.

JSON field	Type / unit	Definition
changes.bid_price_ticks	Array[5] / ticks	Change in each bid price level relative to the preceding snapshot.
changes.bid_volume	Array[5] / shares	Change in displayed bid volume at each level relative to the preceding snapshot.
changes.ask_price_ticks	Array[5] / ticks	Change in each ask price level relative to the preceding snapshot.
changes.ask_volume	Array[5] / shares	Change in displayed ask volume at each level relative to the preceding snapshot.

7. Reproducible history-summary definitions

The following variables are constructed from the 50 snapshots. They are not pre-existing columns in the **Orders** sheet. The definitions below make the project brief's required history summaries explicit and reproducible.

Let snapshot index **j** run from 1 to 50. Define:

- **mid_j** as the snapshot midpoint;
- **spread_j** as the snapshot spread in ticks;
- **imbalance_j** as **five_level_bid_ask_imbalance**;
- **bid_depth_j = sum(bid_volumes_j)**;
- **ask_depth_j = sum(ask_volumes_j)**;
- **micro_gap_j_bps = 10,000 * (microprice_j - mid_j) / mid_j**.

Price movement

Constructed feature	Definition
mid_return_1_50_bps	**10,000 * (mid_50 - mid_1) / mid_1**.
best_bid_change_count	Number of transitions **j = 2,...,50** for which **bid_price_j[1] != bid_price_(j-1)[1]**.
best_ask_change_count	Number of transitions **j = 2,...,50** for which **ask_price_j[1] != ask_price_(j-1)[1]**.
net_best_quote_direction	Sum of signed L1 bid and ask price changes in ticks across transitions 2-50. If side orientation is used, multiply price direction by **+1** for buy and **-1** for sell and document that convention.

Spread behavior

Constructed feature	Definition
spread_mean_50	Arithmetic mean of **spread_j** over the 50 snapshots.
spread_max_50	Maximum value of **spread_j**.
spread_final	**spread_50**.
spread_one_tick_fraction	Fraction of snapshots for which **spread_j == 1**.

Imbalance behavior

Constructed feature	Definition
imbalance_mean_50	Arithmetic mean of **imbalance_j**.
imbalance_std_50	Population standard deviation of **imbalance_j**, using denominator 50 (**ddof = 0**).
imbalance_min_50	Minimum value of **imbalance_j**.
imbalance_max_50	Maximum value of **imbalance_j**.
imbalance_final	**imbalance_50**.
imbalance_trend_50	Ordinary-least-squares slope from regressing **imbalance_j** on **sequence_step j**.

Depth change

Constructed feature	Definition
bid_depth_change_1_50	**bid_depth_50 - bid_depth_1**.
ask_depth_change_1_50	**ask_depth_50 - ask_depth_1**.
same_side_depth_change_1_50	For buy orders, **bid_depth_50 - bid_depth_1**; for sell orders, **ask_depth_50 - ask_depth_1**.
opposite_side_depth_change_1_50	For buy orders, **ask_depth_50 - ask_depth_1**; for sell orders, **bid_depth_50 - bid_depth_1**.

Microprice signal

Constructed feature	Definition
micro_gap_final_bps	**micro_gap_50_bps**.
micro_gap_mean_50_bps	Arithmetic mean of **micro_gap_j_bps**.
micro_gap_trend_50	Ordinary-least-squares slope from regressing **micro_gap_j_bps** on **sequence_step j**.

Event composition

Constructed feature	Definition
event_count_<type>	Number of snapshots whose **event_type** equals **<type>**, for each of **limit_submission**, **cancellation**, **market_trade**, and **modification**.
event_count_<type>_<side>	Number of snapshots with the specified **event_type** and **event_side**, for bid and ask separately.

Event timing

- **lookback_span_ms**: **event_timestamp_50 - event_timestamp_1**, expressed in milliseconds.
- **mean_inter_event_ms**: arithmetic mean of the 49 consecutive timestamp differences.
- **last_event_to_submission_ms**: **submission_timestamp - event_timestamp_50**, expressed in milliseconds. It should equal 1 ms under the documented schema.